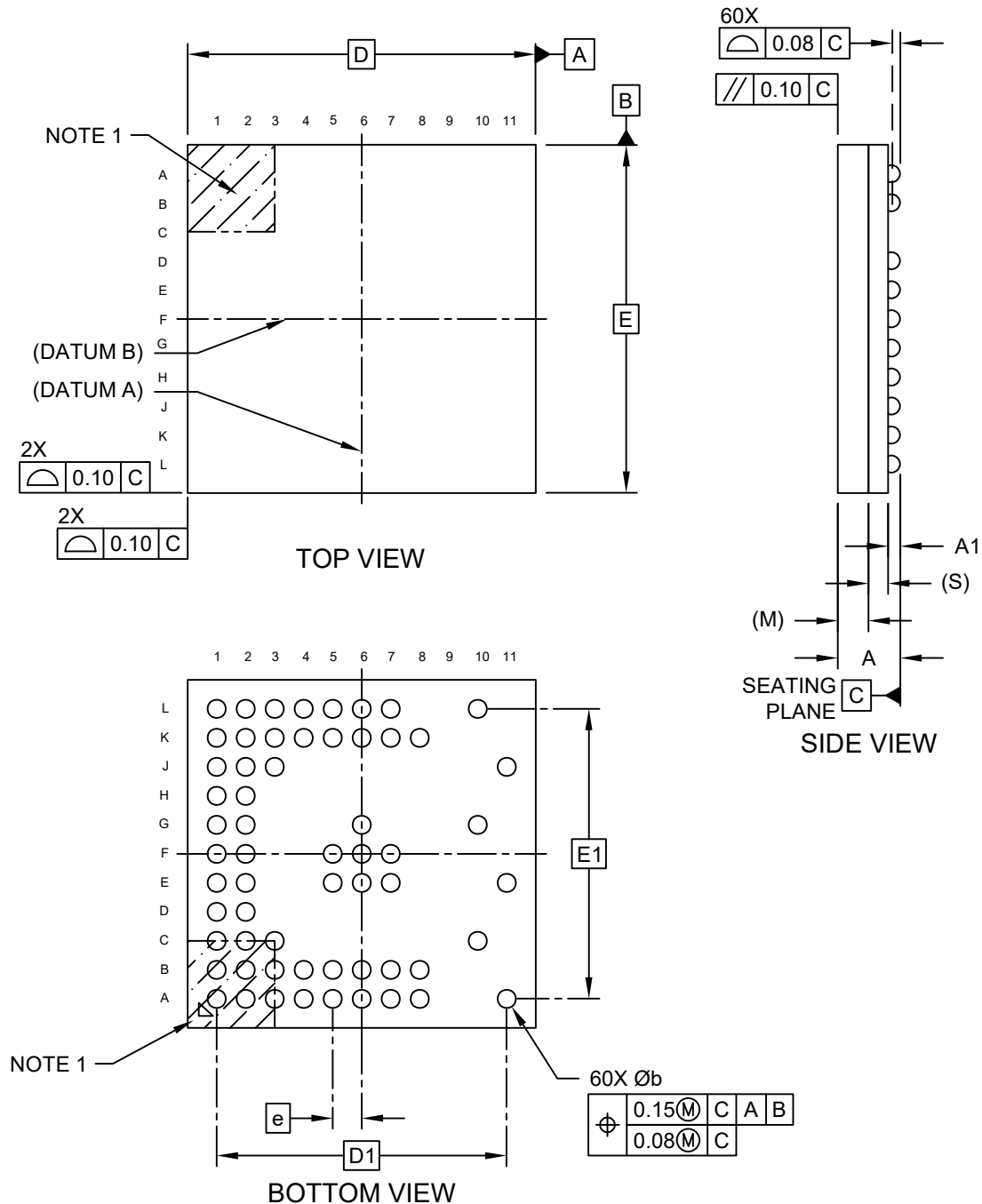


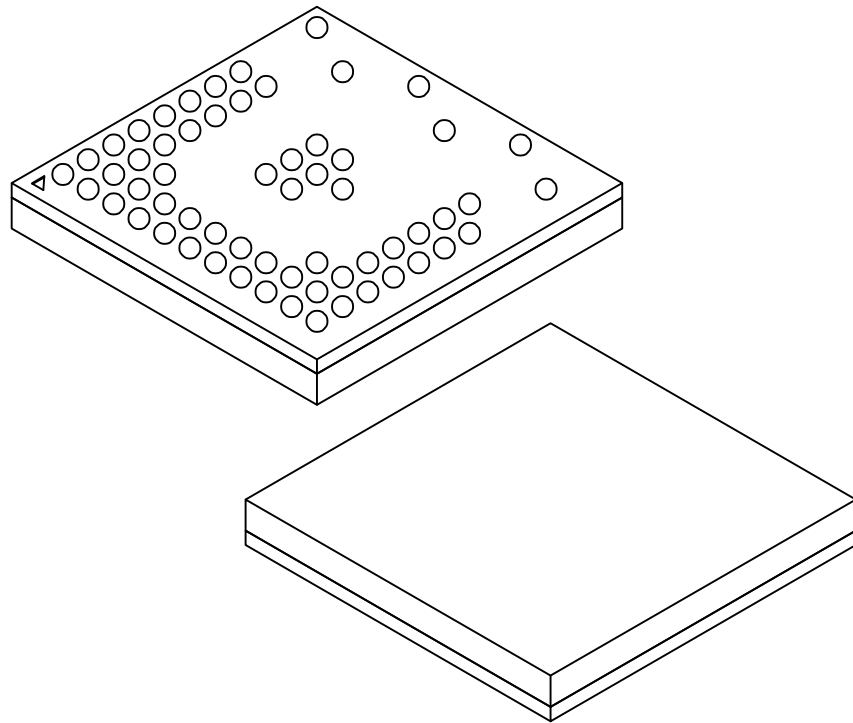
## 60-Ball Thin Fine-Pitch Ball Grid Array (AKA) - 6x6x1.20 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 60-Ball Thin Fine-Pitch Ball Grid Array (AKA) - 6x6x1.20 mm Body [TFBGA]

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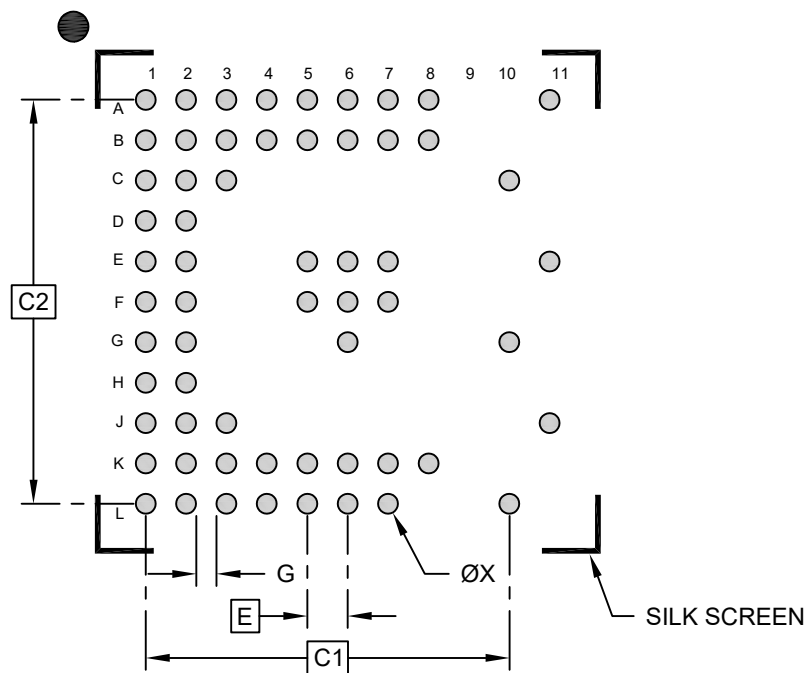
| Units               |    | MILLIMETERS |      |      |
|---------------------|----|-------------|------|------|
| Dimension Limits    |    | MIN         | NOM  | MAX  |
| Number of Terminals | N  | 60          |      |      |
| Pitch               | e  | 0.50 BSC    |      |      |
| Overall Height      | A  | –           | –    | 1.20 |
| Ball Height         | A1 | 0.16        | –    | 0.26 |
| Mold Thickness      | M  | 0.53 REF    |      |      |
| Substrate Thickness | S  | 0.248 REF   |      |      |
| Overall Length      | D  | 6.00 BSC    |      |      |
| Ball Array Length   | D1 | 5.00 BSC    |      |      |
| Overall Width       | E  | 6.00 BSC    |      |      |
| Ball Array Width    | E1 | 5.00 BSC    |      |      |
| Ball Width          | b  | 0.27        | 0.30 | 0.37 |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 60-Ball Thin Fine-Pitch Ball Grid Array (AKA) - 6x6x1.20 mm Body [TFBGA]

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### RECOMMENDED LAND PATTERN

| Dimension Limits           | Units | MILLIMETERS |     |      |
|----------------------------|-------|-------------|-----|------|
|                            |       | MIN         | NOM | MAX  |
| Contact Pitch              | E     | 0.50 BSC    |     |      |
| Contact Pad Spacing        | C1    | 5.00 BSC    |     |      |
| Contact Pad Spacing        | C2    | 5.00 BSC    |     |      |
| Contact Pad Width (X60)    | X     |             |     | 0.25 |
| Space Between Contact Pads | G     | 0.25        |     |      |

#### Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.